CPE134 [Fundamentals of Electrical Circuits]

CAPACITOR AND INDUCTOR TRANSIENT ANALYSIS

LABORATORY EXERCISE 6

Leader: _____

Date: _____ Rating: _____

Members: _____

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NEW MESSAGE

From: **Leah Adams** (SPC Manager)

Subject: **Capacitor and Inductor Transient Analysis**

Good day, _____

Hope your doing well! Heard about your team's exceptional capabilities.

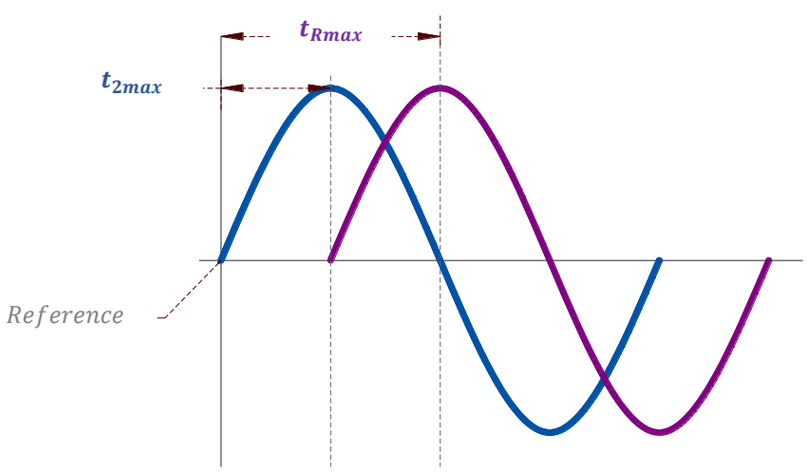
We are currently developing a control chart for the product code: "Lilac". To ensure accuracy and efficiency, we need data of **transient analysis** for its reactive components.

Would it be possible for your team to extend a helping hand in this matter? For more detailed overview of our request, kindly refer to the attached files.

Thank you for your support.

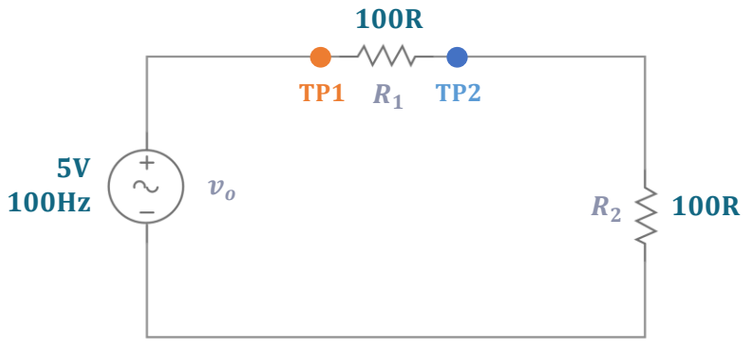
Best Regards,
Leah

Reference Curve,



EXERCISE 1 RESISTOR CHARACTERISTIC CURVE

Step 1 **Set-up** the given circuit below.



Step 2 Perform **MATH → CH1 – CH2** then capture the waveform *[attached]* .

Step 3 Phase Shift Analysis

How long does it take for **CH2** voltage to reach its peak voltage with respect to reference, t_0 ?

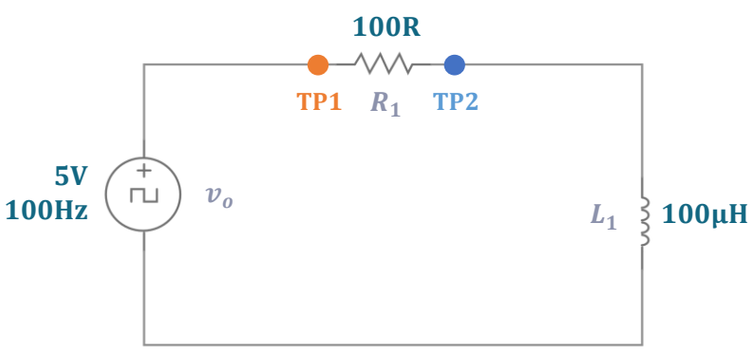
t_{2max} = _____

How long does it take for **MATH RESULT** voltage to reach its peak voltage with respect to reference, t_0 ?

t_{Rmax} = _____

EXERCISE 2 INDUCTOR CHARACTERISTIC CURVE

Step 1 **Set-up** the given circuit below.



Step 2 Perform **MATH → CH1 – CH2** then capture the waveform *[attached]* .

Step 3 Phase Shift Analysis

How long does it take for **CH2** voltage to reach its peak voltage with respect to reference, t_0 ?

t_{2max} = _____

How long does it take for **MATH RESULT** voltage to reach its peak voltage with respect to reference, t_0 ?

t_{Rmax} = _____

Step 4 **Time Constant,τ**

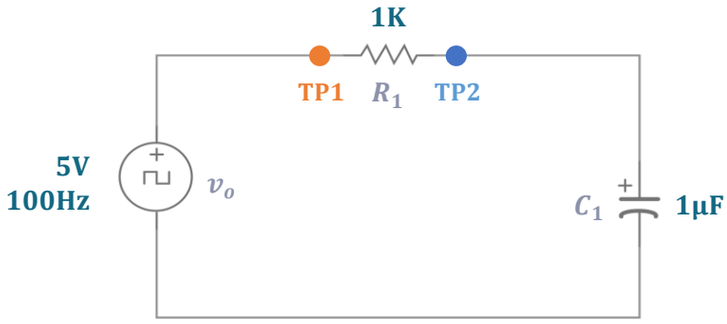
Calculate the time constant of the inductor using the formula:

$$\tau = \frac{L}{R}$$

$$5\tau =$$

EXERCISE 3 **CAPACITOR** CHARACTERISTIC CURVE

Step 1 **Set-up** the given circuit below.



Step 2 Perform **MATH → CH1 – CH2** then capture the waveform *[attached]* .

Step 3 Phase Shift Analysis

How long does it take for **CH2** voltage to reach its peak voltage with respect to reference, t_0 ?

t_{2max} = _____

How long does it take for **MATH RESULT** voltage to reach its peak voltage with respect to reference, t_0 ?

t_{Rmax} = _____

Step 4 **Time Constant,τ**

Calculate the time constant of the capacitor using the formula:

$\tau = RC$ $5\tau =$

Conclusion,

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_____

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